PHP Operators

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PHP Operators

Operators are used to perform operations on variables and values.

PHP divides the operators in the following groups:

- Arithmetic operators
- Assignment operators
- Comparison operators
- Increment/Decrement operators
- Logical operators
- String operators
- Array operators
- Conditional assignment operators

PHP Arithmetic Operators

The PHP arithmetic operators are used with numeric values to perform common arithmetical operations, such as addition, subtraction, multiplication etc.

Operator	Name	Example	Result	Try it
+	Addition	$x + $y	Sum of $x and $y	Try it »

/	Division	$x / $y	Quotient of $x and $y	Try it »
%	Modulus	$x % $y	Remainder of $x divided by $y	Try it »
**	Exponentiation	$x ** $y	Result of raising $x to the $y'th power	Try it »

PHP Assignment Operators

The PHP assignment operators are used with numeric values to write a value to a variable.

The basic assignment operator in PHP is "=". It means that the left operand gets set to the value of the assignment expression on the right.

Assignment	Same as...	Description	Try it
x = y	x = y	The left operand gets set to the value of the expression on the right	Try it »
x += y	x = x + y	Addition	Try it »
x -= y	x = x - y	Subtraction	Try it »
x *= y	x = x * y	Multiplication	Try it »
x /= y	x = x / y	Division	Try it »
x %= y	x = x % y	Modulus	Try it »

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PHP Comparison Operators

The PHP comparison operators are used to compare two values (number or string):

Operator	Name	Example	Result	Try it
==	Equal	$x == $y	Returns true if $x is equal to $y	Try it »
===	Identical	$x === $y	Returns true if $x is equal to $y, and they are of the same type	Try it »
!=	Not equal	$x != $y	Returns true if $x is not equal to $y	Try it »
<>	Not equal	$x <> $y	Returns true if $x is not equal to $y	Try it »
!==	Not identical	$x !== $y	Returns true if $x is not equal to $y, or they are not of the same type	Try it »
>	Greater than	$x > $y	Returns true if $x is greater than $y	Try it »
<	Less than	$x < $y	Returns true if $x is less than $y	Try it »
>=	Greater than or equal to	$x >= $y	Returns true if $x is greater than or equal to $y	Try it »
<=	Less than or equal to	$x <= $y	Returns true if $x is less than or equal to $y	Try it »

		$x is less than, equal to, or greater than $y. Introduced in PHP 7.

PHP Increment / Decrement Operators

The PHP increment operators are used to increment a variable's value.

The PHP decrement operators are used to decrement a variable's value.

Operator	Same as...	Description	Try it
++$x	Pre-increment	Increments $x by one, then returns $x	Try it »
$x++	Post-increment	Returns $x, then increments $x by one	Try it »
--$x	Pre-decrement	Decrements $x by one, then returns $x	Try it »
$x--	Post-decrement	Returns $x, then decrements $x by one	Try it »

PHP Logical Operators

The PHP logical operators are used to combine conditional statements.

Operator	Name	Example	Result	Try it
and	And	$x and $y	True if both $x and $y are true	Try it »
or	Or	$x or $y	True if either $x or $y is true	Try it »
xor	Xor	$x xor $y	True if either $x or $y is true, but not both	Try it »

			true	
!	Not	!$x	True if $x is not true	Try it »

PHP String Operators

PHP has two operators that are specially designed for strings.

Operator	Name	Example	Result	Try it
.	Concatenation	$txt1 . $txt2	Concatenation of $txt1 and $txt2	Try it »
.=	Concatenation assignment	$txt1 .= $txt2	Appends $txt2 to $txt1	Try it »

PHP Array Operators

The PHP array operators are used to compare arrays.

Operator	Name	Example	Result	Try it
+	Union	$x + $y	Union of $x and $y	Try it »
==	Equality	$x == $y	Returns true if $x and $y have the same key/value pairs	Try it »
===	Identity	$x === $y	Returns true if $x and $y have the same key/value pairs in the same order and of the same types	Try it »
!=	Inequality	$x != $y	Returns true if $x is not equal to $y	Try it »

identical to $y

PHP Conditional Assignment Operators

The PHP conditional assignment operators are used to set a value depending on conditions:

Operator	Name	Example	Result	Try it
?:	Ternary	$x = *expr1* ? *expr2* : *expr3*	Returns the value of $x. The value of $x is *expr2* if *expr1* = TRUE. The value of $x is *expr3* if *expr1* = FALSE	Try it »
??	Null coalescing	$x = *expr1* ?? *expr2*	Returns the value of $x. The value of $x is *expr1* if *expr1* exists, and is not NULL. If *expr1* does not exist, or is NULL, the value of $x is *expr2*. Introduced in PHP 7	Try it »

Exercise ?

Which one is a correct syntax for incrementing $x by 1:

- `$x + 1;`
- `$x++;`
- `1 + $x;`